

04 Determinants

4.2 The 3×3 Determinant

Understanding the 3×3 determinant through parallelepiped volume



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills for This Section

1

A 3×3 matrix's determinant represents the signed volume of the parallelepiped formed by its column vectors

3

The identity matrix's columns form a unit cube with volume 1, so its determinant is 1

5

Upper- and lower-triangular matrices' determinants also equal the product of diagonal entries; a zero on the diagonal makes the determinant 0

7

Swapping any two columns or rows flips the handedness of the system, negating the determinant

2

The right-hand rule tells the determinant's sign: a right-handed system gives a positive determinant, a left-handed one negative

4

A diagonal matrix's determinant equals the product of its diagonal entries, matching a cube's or box's volume

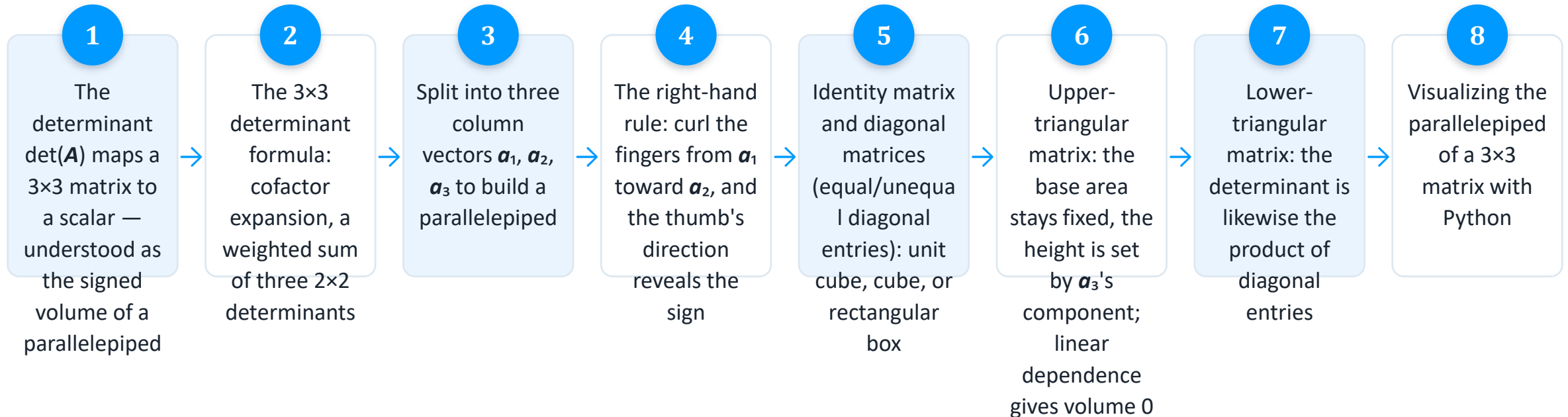
6

If a 3×3 matrix's three columns are coplanar or collinear (linearly dependent), they can't span a volume, so the determinant is 0

8

The determinant is a mapping from an $n \times n$ matrix to a scalar — the area-to-volume pattern generalizes to higher dimensions

Roadmap for This Section



From Parallelogram Area to Parallelepiped Volume

Last section: $\det(\mathbf{A})$ was the signed area of a 2×2 matrix; this section: $\det(\mathbf{A})$ is the signed volume of a 3×3 matrix

- The determinant is only defined for square matrices; for non-square matrices the concept does not apply
- The determinant $\det()$ is a mapping from an $n \times n$ square matrix to a scalar, revealing key properties such as the area/volume scaling factor and invertibility
- This section mirrors the previous one almost step for step — reading them side by side makes the geometric meaning of the determinant across dimensions much more intuitive

Cofactor Expansion: A Weighted Sum of Three 2×2 Determinants

$$\det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

- Given a 3×3 matrix **A**, write its entries as $a_{i,j}$, where i is the row index and j the column index
- M_{1j} is the minor obtained by deleting row 1 and column j — the determinant of the remaining 2×2 submatrix
- For now, don't worry about where this formula comes from — just use it as given; its derivation is covered at the end of this chapter

The Fully Expanded Formula

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

- Working through the three 2×2 determinants gives this six-term expansion, each term a product of three entries
- This pattern generalizes to $n \times n$ matrices — a weighted sum over many terms, the seed of what becomes Laplace expansion
- This section focuses on the geometric picture: how a parallelepiped makes the 3×3 determinant intuitive

Splitting the Matrix into Three Column Vectors

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3] \Rightarrow \text{three edges span a parallelepiped}$$

- Earlier in this book, summing three vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ was shown to construct a parallelepiped
- Here we reuse that same geometric construction to understand the 3×3 determinant: the three column vectors are exactly the parallelepiped's three edges

Figure 1 · Parallelepiped

- (a) $\mathbf{a}_1$ (red), $\mathbf{a}_2$ (green), $\mathbf{a}_3$ (blue) start at the origin; the black diagonal is their sum $\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3$
- (b) Right-hand rule: curl the fingers from $\mathbf{a}_1$ toward $\mathbf{a}_2$ — if the thumb points the same way as $\mathbf{a}_3$, it's right-handed and the determinant is positive

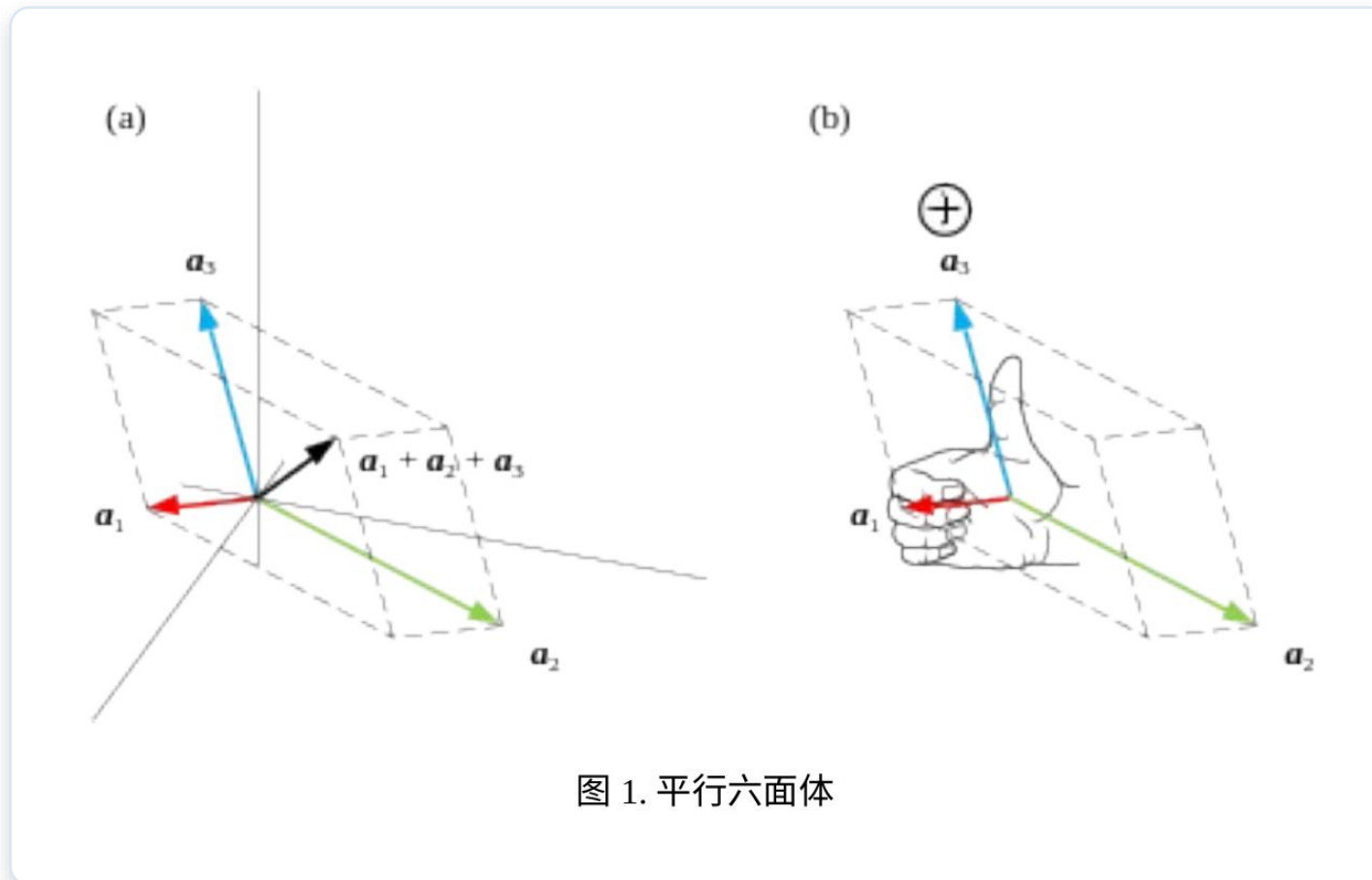


图 1. 平行六面体

The Right-Hand Rule for the Determinant's Sign

- 1) Make a fist with your right hand, and curl your four fingers
- 2) The direction your fingers point represents matrix $\mathbf{A}$'s first column vector $\mathbf{a}_1$
- 3) Your palm faces the second column vector $\mathbf{a}_2$
- 4) Your extended thumb points along the third column vector $\mathbf{a}_3$

Conclusion If the thumb's direction matches $\mathbf{a}_3$, the determinant is positive (right-handed); if opposite, it's negative (left-handed)

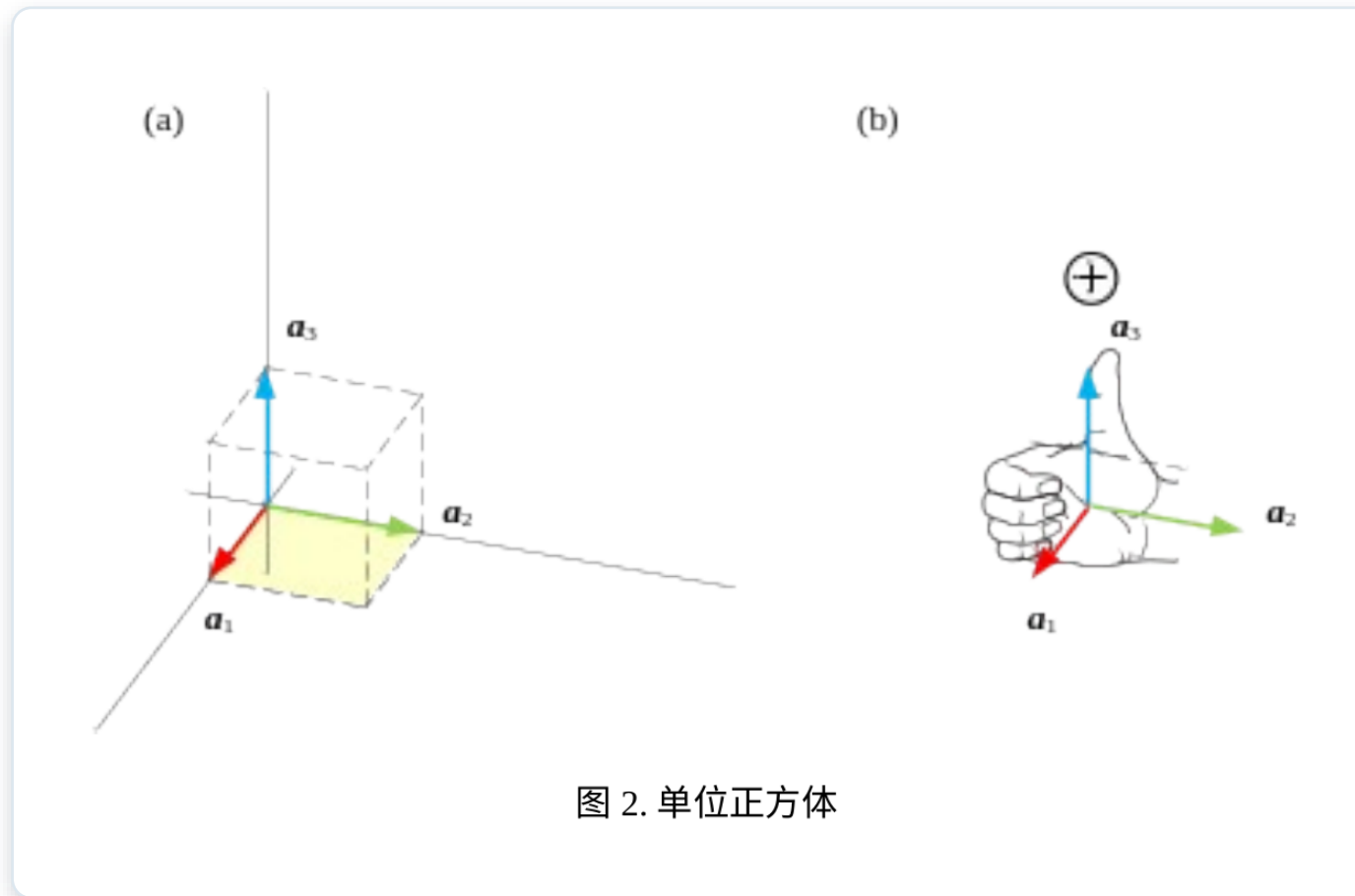
The Identity Matrix Has Determinant 1

$$A = [1 \ 0 \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1] \Rightarrow \det(A) = 1$$

- The identity matrix is a special diagonal matrix, so its determinant is the product of its diagonal entries: $1 \times 1 \times 1 = 1$
- $\mathbf{a}_1$ and $\mathbf{a}_2$ form a unit-square base with area 1; $\mathbf{a}_3$ is the height, also 1, so the volume (determinant) is 1
- $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ satisfy the right-hand rule, so the determinant is positive

Figure 2 · Unit Cube

- (a) The three column vectors $\mathbf{a}_1=[1;0;0]$, $\mathbf{a}_2=[0;1;0]$, $\mathbf{a}_3=[0;0;1]$ form a unit cube with edge length 1
- (b) Right-hand rule check: the thumb points the same way as $\mathbf{a}_3$, so the determinant is positive



Swapping Columns 1 and 2 Makes the Determinant -1

$$A = [0 \ 1 \ 0; 1 \ 0 \ 0; 0 \ 0 \ 1] \Rightarrow \det(A) = -1$$

- After swapping columns 1 and 2 of the identity matrix, the column vectors still form a unit cube of volume 1
- But now the thumb points opposite to $\mathbf{a}_3$ — the columns form a left-handed system, so the determinant becomes -1
- This shows swapping any two columns flips the handedness and negates the determinant — the 3D analogue of an orientation flip

Figure 3 · Unit Cube, Columns 1 and 2 Swapped

- (a) After the swap the column vectors still span a unit cube — volume still 1
- (b) But $\mathbf{a}_3$'s direction is opposite the thumb, showing a left-handed system — the determinant is negative

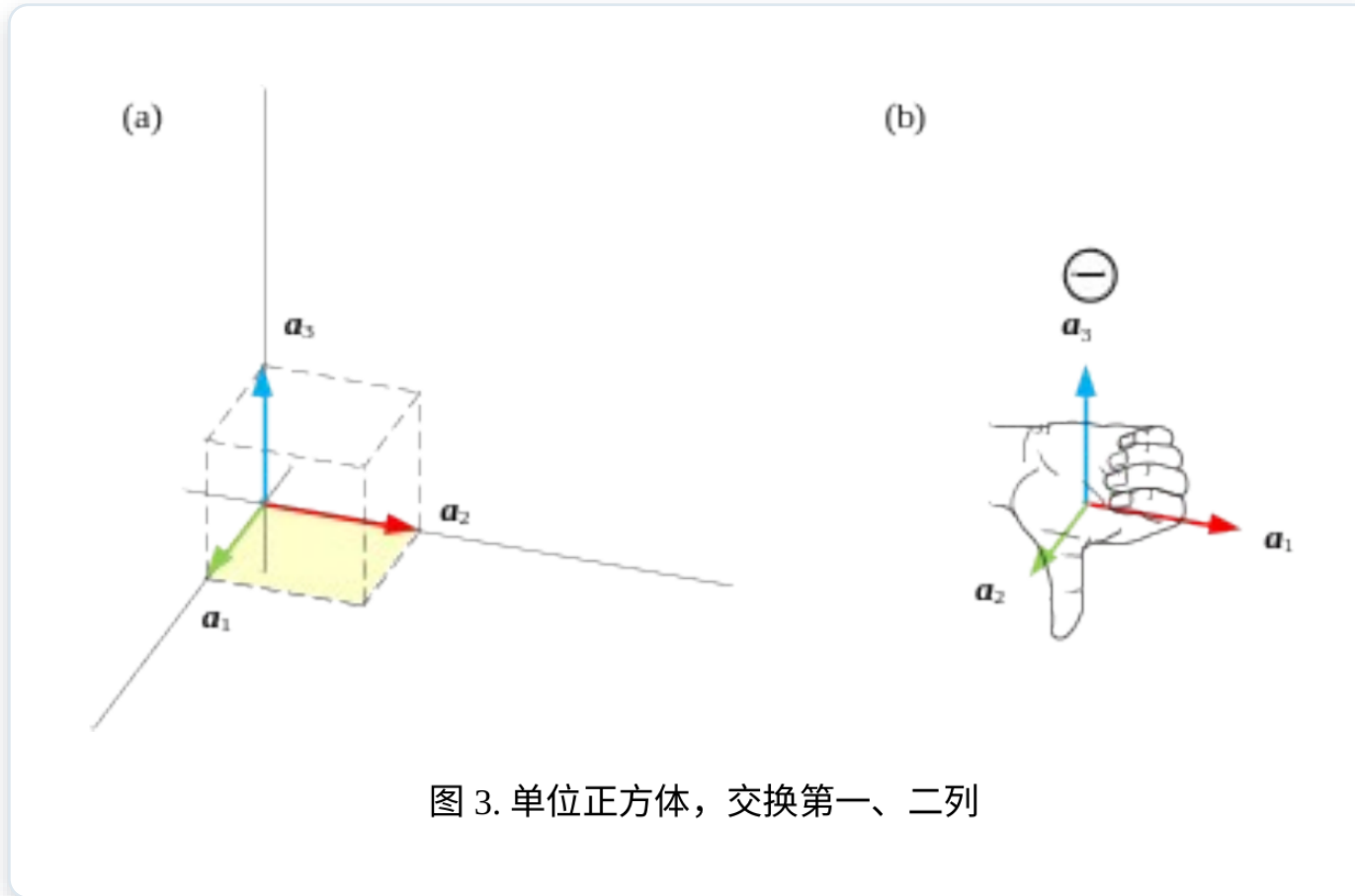


图 3. 单位正方体，交换第一、二列

A Cube with Equal Edge Lengths

$$A = [3 \ 0 \ 0; 0 \ 3 \ 0; 0 \ 0 \ 3] \Rightarrow \det(A) = 3 \times 3 \times 3 = 27$$

- A diagonal matrix's determinant equals the product of its diagonal entries
- The three column vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ span a cube with edge length 3, giving a volume of 27
- The column vectors satisfy the right-hand rule, so the determinant is positive

Figure 4 · A Cube

- $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ point in different directions but share the same length (3), forming a cube
- Analyze on your own how the determinant of this kind of equal-diagonal 3×3 matrix relates to its edge length

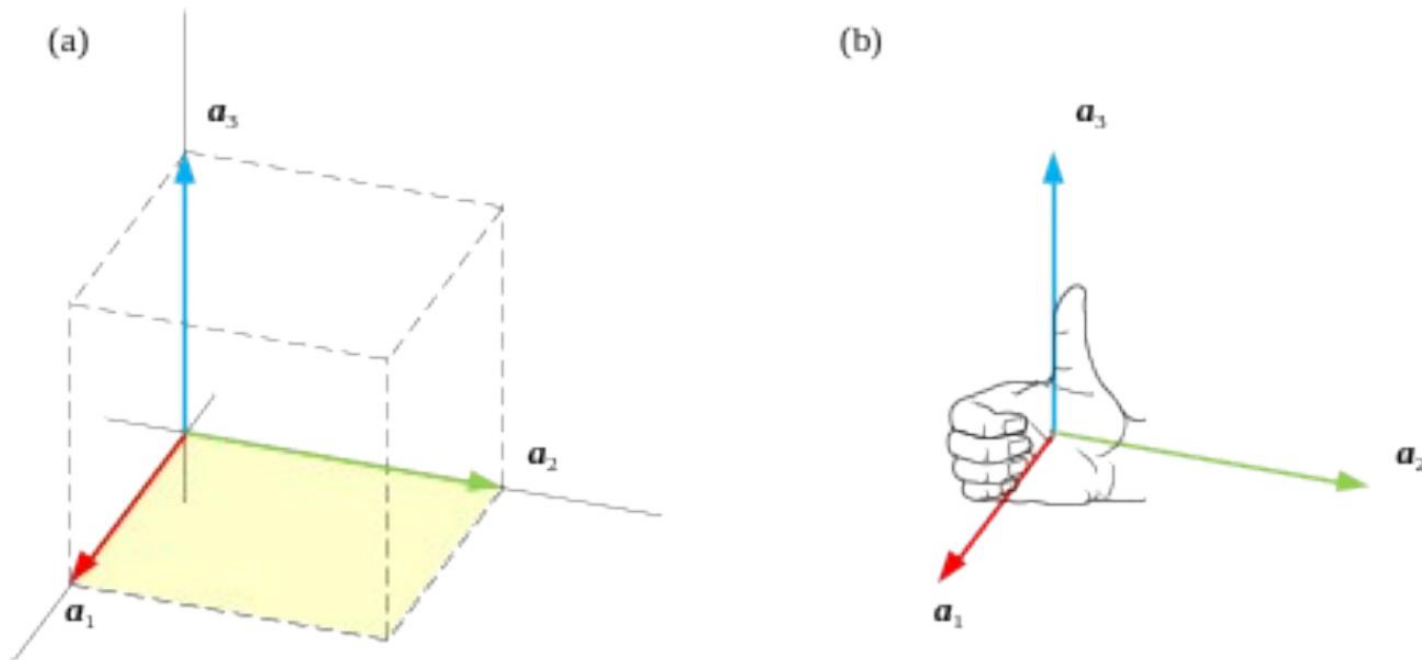


图 4. 正方体

Length × Width × Height = Box Volume

$$A = [3 \ 0 \ 0; 0 \ 2 \ 0; 0 \ 0 \ 1] \Rightarrow \det(A) = 3 \times 2 \times 1 = 6$$

- When a diagonal matrix's entries are unequal, A 's geometric shape is a rectangular box
- The determinant is exactly “length × width × height” — the product of the three diagonal entries
- a_1, a_2, a_3 satisfy the right-hand rule, so the determinant is positive

Figure 5 · A Rectangular Box

- Three mutually perpendicular edges of different lengths form a box, whose volume is the product of the three edge lengths
- Right-hand rule check: $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ satisfy the right-hand rule, so the determinant is positive

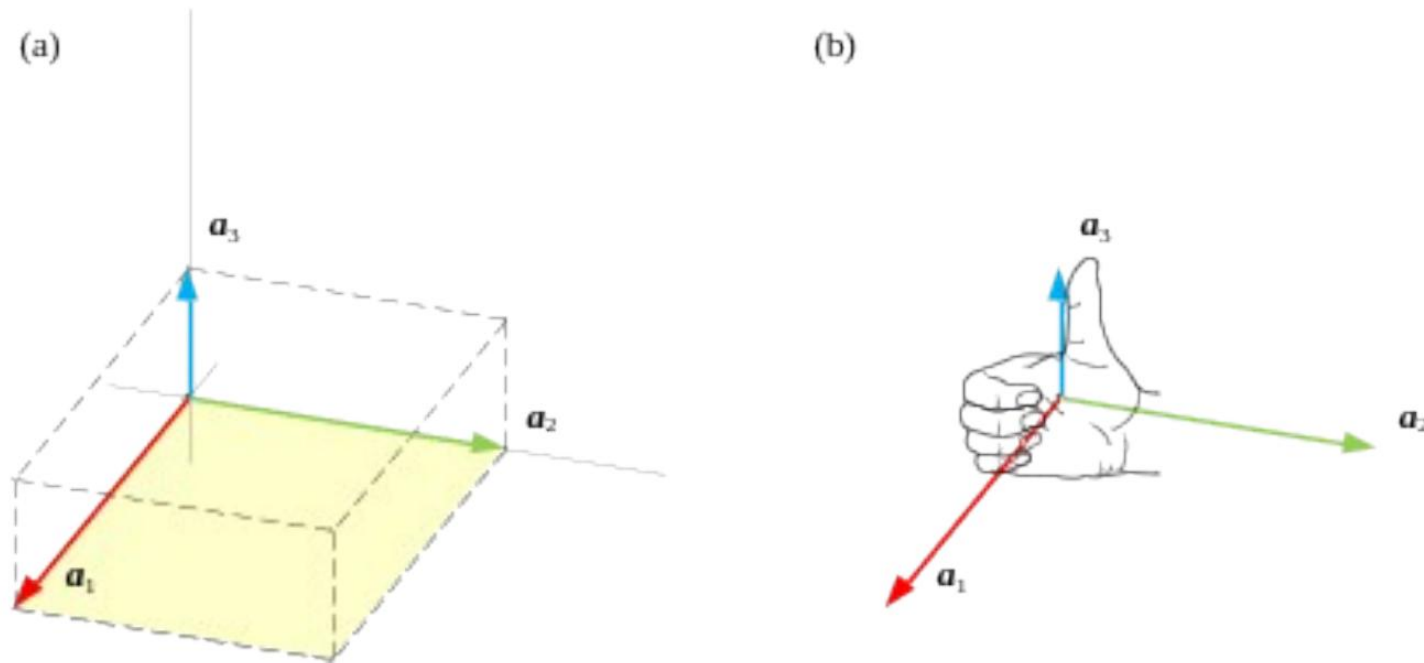


图 5. 长方体

A Diagonal Matrix's Determinant Is the Product of Its Diagonal

$$\text{diagonal matrix } \text{diag}(d_1, d_2, d_3) \Rightarrow \det(\mathbf{A}) = d_1 \times d_2 \times d_3$$

- This generalizes to any $n \times n$ diagonal matrix: $\det(\mathbf{A}) = d_1 \times d_2 \times \cdots \times d_n$
- If any diagonal entry is 0, the determinant is 0 — the corresponding “volume” collapses into a lower-dimensional shape

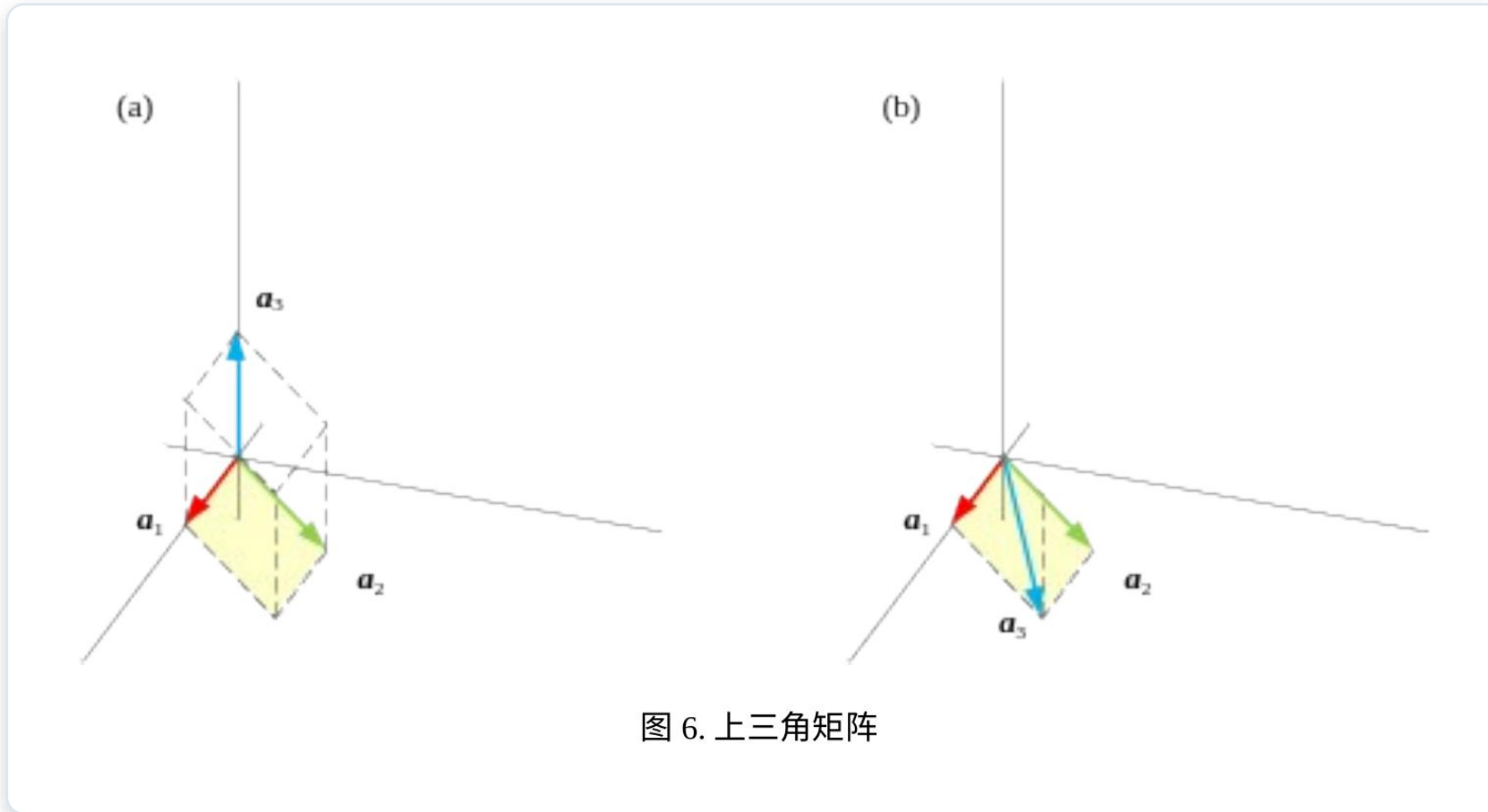
Horizontal Shear, Volume Unchanged

$$A = [1 \ 1 \ 0; 0 \ 1 \ 0; 0 \ 0 \ 1] \Rightarrow \det(A) = 1 \times 1 \times 1 = 1$$

- The previous section showed an upper-triangular matrix's determinant equals the product of its diagonal entries
- $\mathbf{a}_1$ lies along the x_1 axis, and $\mathbf{a}_2$ lies in the x_1x_2 plane; together they form a base of area 1. The parallelepiped's height comes only from $\mathbf{a}_3$'s component along x_3 , which is 1
- If $\mathbf{a}_3$ becomes $[2; 1; 0]$, it falls into the plane spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$ — the three vectors become linearly dependent, the parallelepiped “collapses” onto a plane, and the volume (determinant) becomes 0

Figure 6 · Upper-Triangular Matrix

- (a) $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ span a parallelepiped of volume 1, matching the product of diagonal entries
- (b) With $\mathbf{a}_3$ changed to $[2;1;0]$, $\mathbf{a}_3, \mathbf{a}_1, \mathbf{a}_2$ become linearly dependent — the parallelepiped collapses onto a plane, and the determinant is 0



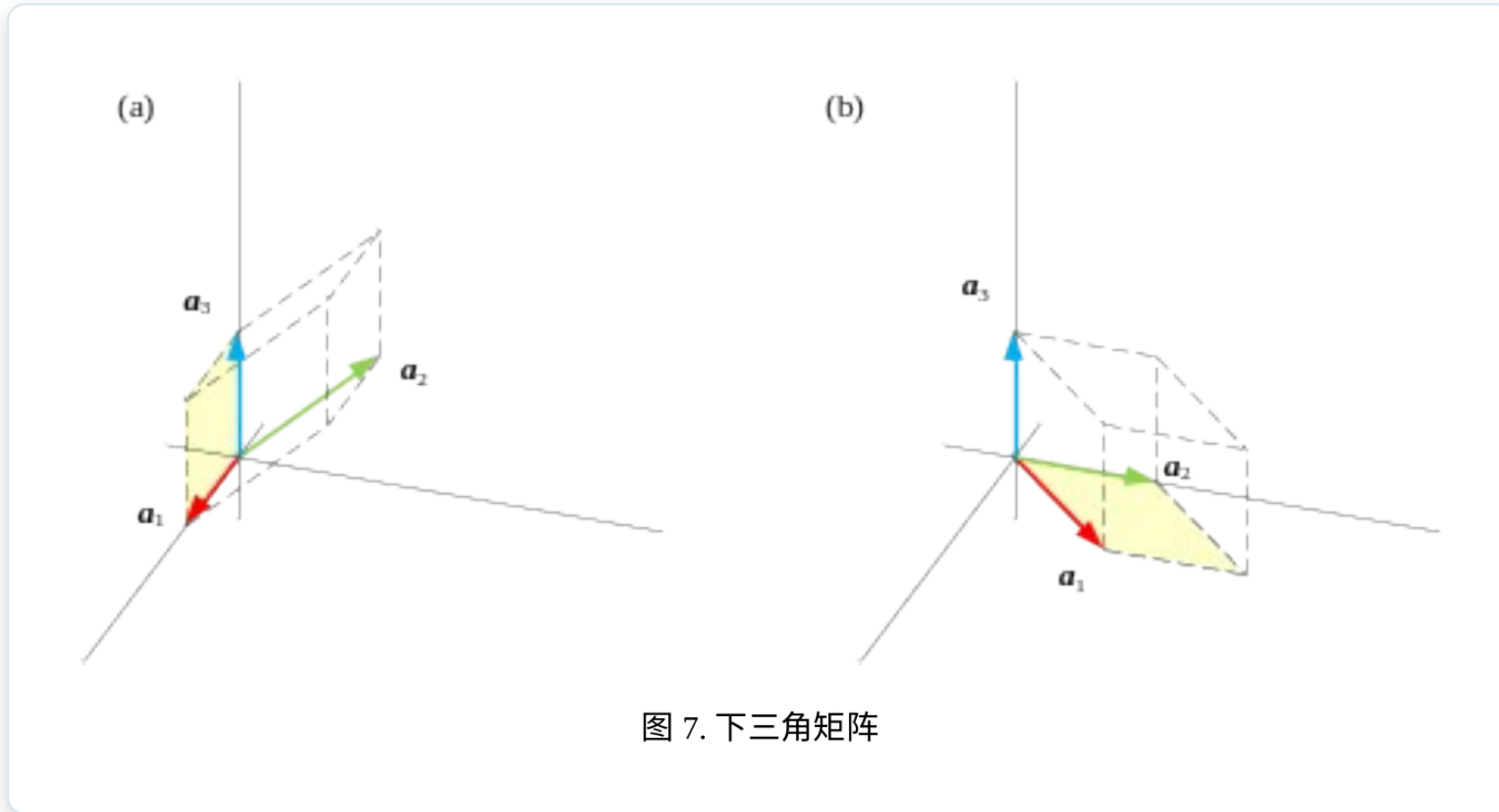
Vertical Shear, Volume Unchanged

$$A = [1 \ 0 \ 0; 0 \ 1 \ 0; 0 \ 1 \ 1] \Rightarrow \det(A) = 1 \times 1 \times 1 = 1$$

- Like diagonal and upper-triangular matrices, a lower-triangular matrix's determinant is also the product of its diagonal entries
- In the example of Figure 7(a), the three column vectors span a parallelepiped of volume 1
- Figure 7(b) shows a second lower-triangular example $[1,0,0;1,1,0;0,0,1]$, whose determinant is also 1 — analyze its parallelepiped shape on your own

Figure 7 · Lower-Triangular Matrix

- (a)(b) Two lower-triangular examples, both spanning parallelepipeds of volume 1
- Verify for yourself: however the shape gets sheared, the volume (determinant) stays the same as long as the diagonal entries are unchanged



A Triangular Matrix's Determinant Is Also the Product of Its Diagonal

upper (or lower) triangular, with diagonal entries $d_1, d_2, d_3 \Rightarrow \det(A) = d_1 \times d_2 \times d_3$

- An upper-triangular matrix's determinant is decided purely by its diagonal entries, since they alone determine the parallelepiped's base area and vertical height
- Like diagonal matrices, a lower-triangular matrix's determinant follows the same rule: if any diagonal entry is 0, the determinant is 0

Visualizing a Matrix's Columns and Parallelepiped in Python

First extract the matrix's three column vectors, build the parallelepiped's eight vertices and six faces, then draw with `quiver()` and `Poly3DCollection`

- `A[:,0]`, `A[:,1]`, `A[:,2]` extract A's three columns into `a1`, `a2`, `a3`
- The 8 vertices come from the origin, the three column vectors, their pairwise sums, and their triple sum; the 6 faces each consist of 4 vertices
- `ax.quiver()` draws the three edge arrows; `Poly3DCollection` fills the parallelepiped's six faces in semi-transparent yellow with dashed black edges

VISUALIZING A 3×3 MATRIX

Code 1 · Visualizing the Parallelepiped of a 3×3 Matrix

LA_04_02_01.ipynb

- a: Import the Poly3DCollection class from `mpl_toolkits.mplot3d.art3d`, used to draw the faces of a 3D solid
- b: Create a 3×3 matrix $A = \text{np.array}([[2,0,2], [2,2,0], [0,2,2]])$
- c: Extract the three column vectors a_1, a_2, a_3 with $A[:,0], A[:,1], A[:,2]$
- d-e: Build the 8 vertices from pairwise and triple sums of the column vectors, and define the six 4-vertex faces of the parallelepiped
- f-g: Create a 3D subplot with `fig.add_subplot(111, projection='3d')`, then use `ax.quiver()` to draw a_1 (red), a_2 (green), a_3 (blue)
- h: Build the solid with `Poly3DCollection` (alpha=0.3 yellow fill, dashed black edges), then add it to the plot with `ax.add_collection3d()`

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d.art3d import Poly3DCollection

## 定义一个 3×3 矩阵 A
A = np.array([ [2, 0, 2],
               [2, 2, 0],
               [0, 2, 2] ])

## 提取列向量
a1 = A[:, 0]; a2 = A[:, 1]; a3 = A[:, 2]

## 构造平行六面体的8个顶点
O = np.array([0, 0, 0])
P1 = a1; P2 = a2; P3 = a3
P4 = a1 + a2; P5 = a1 + a3; P6 = a2 + a3
P7 = a1 + a2 + a3
vertices = [O, P1, P2, P4, P3, P5, P6, P7]

## 定义六个面
# 每个面是4个点
faces = [[O, P1, P4, P2], # 底面
         [O, P2, P6, P3], # 侧面
         [O, P1, P5, P3], # 侧面
         [P7, P4, P1, P5], # 上面
         [P7, P4, P2, P6], # 侧面
         [P7, P5, P3, P6]] # 背面

## 可视化
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

# 绘制向量箭头
ax.quiver(0, 0, 0, *a1, color='red', linewidth=2)
ax.quiver(0, 0, 0, *a2, color='green', linewidth=2)
ax.quiver(0, 0, 0, *a3, color='blue', linewidth=2)

# 绘制平行六面体
face_collection = Poly3DCollection(faces, alpha=0.3,
                                   facecolor='yellow',
                                   edgecolor='black', ls='--')
ax.add_collection3d(face_collection)

# 图像装饰
ax.set_xlim([0, 3]); ax.set_ylim([0, 3]); ax.set_zlim([0, 3])
ax.set_xlabel("x1"); ax.set_ylabel("x2"); ax.set_zlabel("x3")
ax.view_init(elev=30, azim=30) # 设置视角
ax.set_proj_type('ortho')
ax.set_box_aspect([1,1,1])
```

代码 1. 可视化 3 × 3 方阵对应的平行六面体 | LA_04_02_01.ipynb

The Determinant: A Signed Ruler from 2D Area to 3D Volume

$\det(A)$ = the parallelepiped's signed volume — the geometric meaning of a 3×3 matrix's determinant

- Identity $\rightarrow$ unit cube; diagonal $\rightarrow$ cube or box; upper-/lower-triangular $\rightarrow$ a sheared parallelepiped with unchanged volume — all of these have determinants equal to the product of diagonal entries
- The determinant's sign is set by whether the column vectors form a right-handed system (the right-hand rule); when the three columns are coplanar or collinear (linearly dependent), the determinant is 0 and the matrix is not invertible
- The pattern from 2×2 area to 3×3 volume generalizes to the “hyper-volume” of an $n \times n$ matrix — a theme this chapter continues to explore

Key Terms in This Section

Determinant $\det(A)$ — defined only for square matrices, always a scalar

Parallelepiped the solid spanned by three 3D column vectors; its volume's absolute value equals the determinant's absolute value

Right-/Left-Handed System the column vectors' arrangement: right-handed gives a positive determinant, left-handed a negative one

Determinant of a Diagonal Matrix equals the product of the main-diagonal entries: $d_1 \times d_2 \times d_3$

Linear Dependence when three column vectors are coplanar or collinear they are linearly dependent, cannot span a volume, and $\det = 0$

3×3 Determinant Formula cofactor expansion into a weighted sum of three 2×2 determinants, six terms total

Right-Hand Rule curl fingers from a_1 toward a_2 ; if the thumb matches a_3 's direction, it's right-handed and $\det$ is positive

Unit Cube the cube formed by the identity matrix's three column vectors, with volume 1

Determinant of a Triangular Matrix all entries below (or above) the main diagonal are irrelevant — it too equals the product of the diagonal

Singular Matrix a square matrix with determinant 0 — not invertible

GO FURTHER

Companion Code and Further Reading

Further Reading · LA_04_02_01.ipynb

Visualize the parallelepiped of a 3×3 matrix; practice building vertices and faces

Further Reading · Determinants and Linear Dependence

When three column vectors are linearly dependent, the determinant is necessarily 0 — a key tool for testing linear independence

Coming Up · Next Section

4.3 Properties of Determinants: how determinants behave under row and column operations, plus more computation techniques

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Questions Q1 – Q8

Q1 Compute the determinant of each diagonal matrix below, and describe the shape and orientation of the parallelepiped formed by its column vectors, using the right-hand rule to judge the sign: ① $\text{diag}(1, -1, 1)$; ② $\text{diag}(1, -1, -1)$; ③ $\text{diag}(1, 2, 3)$; ④ $\text{diag}(1, 0, 3)$ (blank entries are 0).

Q3 Compute the determinant of each lower-triangular matrix below, describe the shape of the parallelepiped, and use the right-hand rule to judge the sign: ① $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 3 \end{bmatrix}$.

Q5 Write a Python function `det_3x3(A)` that manually implements the six-term expansion for a 3×3 determinant, and compare it against `np.linalg.det(A)`.

Q7 Explore the connection between the determinant and the scalar triple product: verify whether $\det([a_1, a_2, a_3])$ equals $a_1 \cdot (a_2 \times a_3)$.

Q2 Compute the determinant of each upper-triangular matrix below, describe the shape of the parallelepiped, and use the right-hand rule to judge the sign: ① $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$.

Q4 Swap any one pair of column vectors in the matrices from Q1, Q2, and Q3, and recompute the determinant.

Q6 Write a function `parallelepiped_volume(A)` that returns the volume of the parallelepiped formed by A's three column vectors (the absolute value of the determinant).

Q8 Construct a 3×3 matrix whose columns are linearly dependent (e.g., the third column equals the sum of the first two), compute its determinant, and explain why it is 0 in terms of the parallelepiped “collapsing” onto a plane.

Questions Q9 – Q16

Q9 Write a function `swap_columns_det_3x3(A)` that swaps any two columns of A and recomputes the determinant, verifying it is always the negative of the original.

Q11 Write a function `add_column_multiple_3x3(A, k)` that adds k times A 's first column to its second column, verifying whether the determinant stays unchanged, and explain why.

Q13 Write a function `triangular_det_3x3(A)` that checks whether A is upper- or lower-triangular; if so, return the product of its diagonal directly, otherwise call `np.linalg.det()`.

Q15 Modify the matrix A in Code 1 to construct a 3×3 matrix with linearly dependent columns, and verify whether the plotted “parallelepiped” degenerates into a flat, planar figure.

Q10 Write a function `scale_column_det_3x3(A, k, col)` that multiplies A 's col -th column by scalar k and recomputes the determinant; explore how the determinant changes with k .

Q12 Construct a 3×3 permutation matrix (swap any two columns of the identity matrix), compute its determinant, and cross-check against your finding in Q9.

Q14 Using a Python animation (or a series of subplots), show: fixing a_1 and a_2 , move a_3 continuously out of the plane spanned by a_1 and a_2 , and observe how the parallelepiped's volume (the determinant) changes continuously from 0.

Q16 Write a function `is_invertible_3x3(A)` that returns whether A is invertible by checking whether its determinant is zero (using a tolerance such as $1e-9$ to avoid floating-point error).

Questions Q17 – Q24

Q17 Construct 10 random 3×3 matrices and compare your `det_3x3` function against `np.linalg.det()` numerically, discussing the discrepancies floating-point computation might introduce.

Q19 Explore: given the identity matrix I , construct kI (k a scalar), compute its determinant, and conjecture the relationship between $\det(kI)$, k , and the matrix's dimension — verify with Python (it should be k^3 in the 3×3 case).

Q21 Write a function `random_singular_matrix_3x3()` that randomly generates a singular (determinant-zero) 3×3 matrix, and visualize why its column vectors are coplanar.

Q23 Explore the connection between the determinant and the cross product / triple product (echoing Q7): use Python to verify whether the determinant of the matrix with columns a_1 , a_2 , a_3 equals the numerical value of $a_1 \cdot (a_2 \times a_3)$.

Q18 Write a function `visualize_det_sign_3d(A)` that draws the parallelepiped formed by A 's column vectors in different colors (e.g., blue for positive, red for negative) depending on the determinant's sign.

Q20 Building on this section, write a function `shear_matrix_3d(k, i, j)` that generates a 3D shear matrix (1s on the diagonal, k at row i , column j), and verify its determinant is always 1.

Q22 Explore: construct one upper-triangular matrix and one lower-triangular matrix, and verify whether their product is still triangular, and whether its determinant still equals the product of the two original determinants.

Q24 Building on this section, write a short summary (comments are fine) covering how the determinant is computed for the identity, diagonal, upper-triangular, and lower-triangular matrices in the 3×3 case, and what each means geometrically.

10 Prompts to Explore Further (1/2)

1

Explain why a 3×3 matrix's determinant can be understood as the signed volume of a parallelepiped.

2

Explain how the right-hand rule is used to judge the sign of a 3×3 determinant, and how it relates to the “counterclockwise is positive” rule in the 2D case.

3

Explain why swapping any two columns of a matrix flips the sign of the determinant, and what kind of orientation flip this corresponds to in 3D space.

4

Explain why diagonal, upper-triangular, and lower-triangular 3×3 matrices all have determinants equal to the product of their main-diagonal entries.

5

Explain the equivalence between the determinant and the scalar triple product $\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$.

10 Prompts to Explore Further (2/2)

6

Explain why the determinant must be 0 when three column vectors are linearly dependent (coplanar or collinear).

7

Explain the relationship between the determinant and whether a matrix is invertible — why is a singular matrix not invertible?

8

Explain how the six-term expansion of the 3×3 determinant is derived from the weighted sum of three 2×2 minors.

9

Explain how the “dimension-stepping” pattern from 2×2 determinants (area) to 3×3 determinants (volume) generalizes to $n \times n$ matrices.

10

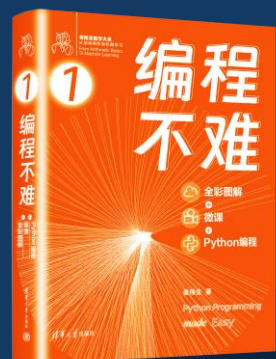
Building on this section, discuss why understanding the geometric meaning of the 3×3 determinant is a foundation for learning the properties of determinants, inverse matrices, and eigenvalues.

数学

数学基础

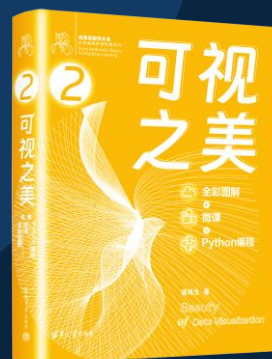
线性代数

概率统计

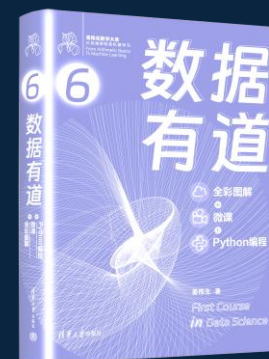
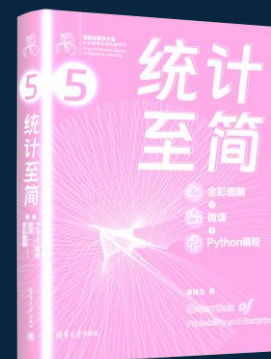
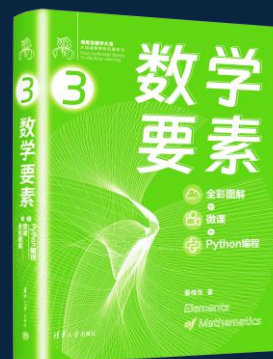


Python编程

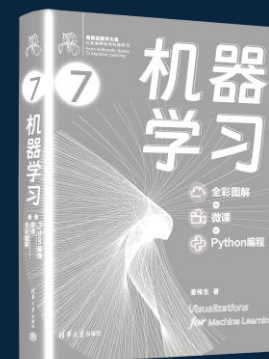
工具



数据可视化



数据分析
图和网络



回归、降维
分类、聚类

实践

